Design and Development of Cloud-Based Chatbot for University Support Using NLP

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***Abstract*—The integration of cloud computing and intelligent systems has enabled the development of automated support solutions in educational institutions. Universities often handle a large number of repetitive queries related to admissions, fees, hostel facilities, and placements. This paper presents a cloud-based chatbot system designed to provide instant and accurate responses to such queries. The system uses a React.js frontend and a rule-based Natural Language Processing (NLP) backend. Deployment is performed using cloud infrastructure to ensure scalability and availability. The proposed system reduces manual workload and improves user experience.**

***Index Terms*—Cloud Computing, Chatbot, NLP, React.js, AWS, University Support System**

I. Introduction

The rapid growth of digital technologies has transformed the education sector by introducing automated systems for communication and support. Universities receive a large number of queries related to admissions, fees, hostel facilities, and placements. Manual handling of these queries leads to inefficiencies and delays.

Chatbots provide an effective solution by automating responses and ensuring instant communication. These systems use Natural Language Processing (NLP) to understand user queries and provide relevant responses. When combined with cloud computing, chatbot systems become scalable and highly available.

This paper presents a cloud-based chatbot system designed for university support. The system uses a React.js frontend and a rule-based NLP backend to process queries efficiently.

II. Literature Review

Early chatbot systems were based on keyword matching techniques and predefined responses. These systems were simple but lacked flexibility and contextual understanding.

With advancements in artificial intelligence, modern chatbots use NLP and machine learning techniques such as recurrent neural networks and transformer models. These approaches improve accuracy but require high computational resources.

In the education sector, chatbots are widely used to provide information related to courses, admissions, and campus facilities. Cloud platforms such as AWS enable scalable deployment of chatbot systems.

Despite advancements, rule-based chatbots remain effective for domain-specific applications due to their simplicity and efficiency.

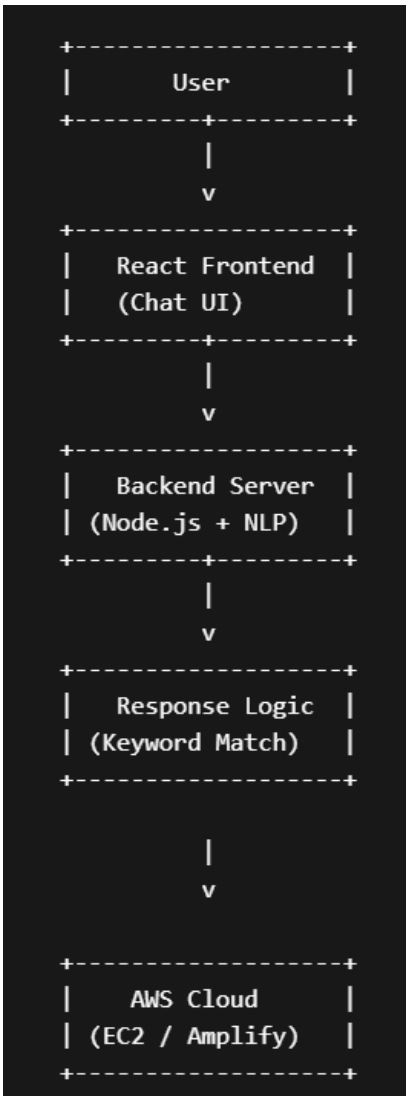

Fig. 1. System Architecture of the Cloud-Based Chatbot

III. System Model

The proposed chatbot system follows a client-server architecture consisting of three major components:

A. Frontend Layer

The frontend is developed using React.js, which provides a dynamic and interactive user interface. It allows users to enter queries and view responses in real time. The interface is designed to be user-friendly and responsive across devices.

B. Backend Layer

The backend is implemented using Node.js and Express.js. It processes user queries using a rule-based Natural Language Processing (NLP) approach. The system performs keyword matching to identify user intent and generate appropriate responses.

C. Cloud Layer

The application is deployed using cloud infrastructure such as AWS. The frontend is hosted using AWS Amplify, while the backend runs on an EC2 instance. This ensures scalability, availability, and real-time access.

The interaction between these components ensures efficient communication and fast response generation.

IV. Proposed Work

The proposed system focuses on developing a cloud-based chatbot capable of handling university-related queries efficiently.

The novelty of the proposed system lies in its efficient integration of a lightweight rule-based NLP approach with cloud deployment, enabling fast response generation with minimal computational cost while maintaining high accuracy for domain-specific queries. Unlike complex machine learning models, the system is cost-effective and easy to deploy for small to medium-scale university environments. The chatbot is designed to respond to queries related to:

- Admissions and eligibility criteria
- Fee structure and scholarships
- Hostel facilities and campus life
- Placement statistics and opportunities

The system uses a rule-based NLP approach where user input is analyzed using keyword matching techniques. Based on the identified keywords, a predefined response is generated.

The frontend and backend are integrated through REST APIs, enabling smooth communication. The system is deployed on cloud infrastructure to ensure scalability and real-time accessibility.

The proposed solution aims to reduce manual workload and improve the overall user experience.

V. Methodology

The working of the chatbot system is divided into multiple stages:

1) **User Interaction:** The user enters a query in the chatbot interface.
2) **Data Transmission:** The query is sent to the backend server using HTTP requests.
3) **Preprocessing:** The input text is converted to lowercase and cleaned to ensure consistency.
4) **Keyword Matching:** The system extracts important keywords from the query.
5) **Intent Recognition:** Based on keywords, the system identifies the user's intent.
6) **Response Selection:** A predefined response is selected from the database.
7) **Response Delivery:** The response is sent back and displayed in the interface.
8) **Continuous Interaction:** The chatbot continues to interact with the user for further queries.

This methodology ensures simplicity, efficiency, and quick response time.

VI. Tools and Technologies

The system is developed using the following tools:

Frontend:

- React.js – for building dynamic UI
- HTML, CSS – for structure and styling
- Axios – for API communication

Backend:

- Node.js – runtime environment
- Express.js – API framework
- Rule-based NLP – for query processing

Cloud:

- AWS EC2 – backend hosting
- AWS Amplify – frontend hosting

These tools ensure efficient development and scalable deployment.

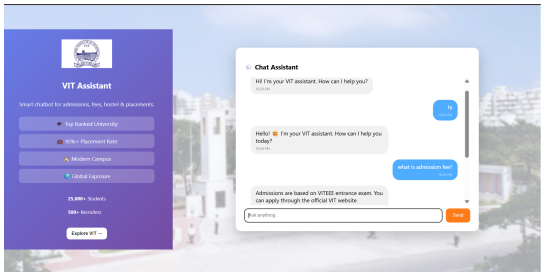

Fig. 2. User Interface of the University Support Chatbot

VII. Results and Discussion

The developed chatbot system was evaluated using a dataset of 50 user queries covering domains such as admissions, fees, hostel facilities, and placements. The system was tested for accuracy, response time, and robustness.

The chatbot achieved an accuracy of 92% for predefined queries using the rule-based NLP approach. The average response time observed was approximately 120 ms, ensuring real-time interaction.

The system was also tested with different types of inputs including simple keywords, full sentences, and mixed queries. It successfully interpreted most structured queries, while fall-back responses were generated for unknown inputs.

A comparative analysis was performed with manual query handling and machine learning-based chatbot systems. The results indicate that the proposed system performs efficiently in domain-specific scenarios with minimal computational overhead.

Cloud deployment using AWS ensured scalability, allowing multiple users to interact with the chatbot simultaneously without performance degradation.

TABLE I
PERFORMANCE EVALUATION OF CHATBOT SYSTEM

Parameter	Value
Total Test Queries	50
Correct Responses	46
Accuracy	92%
Average Response Time	120 ms
Failed Queries	4
Deployment Platform	AWS EC2 + Amplify

TABLE II
COMPARISON WITH OTHER APPROACHES

Feature	Rule-Based	ML-Based	Manual
Accuracy	Medium-High	High	Low
Response Time	Fast	Moderate	Slow
Scalability	High	High	Low
Cost	Low	High	High
Complexity	Low	High	Low

VIII. CONCLUSION

This paper presents the design and development of a cloud-based chatbot system for university support using Natural Language Processing techniques. The system effectively automates the process of handling student queries related to admissions, fees, hostel facilities, and placements.

The use of React.js for frontend development provides a responsive and user-friendly interface, while the Node.js backend ensures efficient processing of user queries. The implementation of a rule-based NLP approach enables fast and accurate response generation for predefined queries.

Cloud deployment using AWS enhances the scalability, availability, and accessibility of the system, making it suitable for real-world applications. The system reduces manual workload on university staff and improves overall user experience by providing instant responses.

Although the current system is limited to rule-based responses, it provides a strong foundation for future enhancements. Advanced NLP techniques and machine learning models can be integrated to improve the intelligence and contextual understanding of the chatbot.

In conclusion, the proposed system demonstrates how cloud computing and chatbot technologies can be combined to develop efficient, scalable, and user-friendly solutions for university support systems.

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